

Review questions

- 1 Explain the difference between:
 - a observations and surveys
 - b longitudinal studies and case studies.
- 2 What is a controlled experiment?
- 3 a What is a hypothesis?
b Can a hypothesis be proved? Explain.
- 4 What is a literature review and what are some of the reasons for carrying out such a review?
- 5 List four principles that must be satisfied if an investigation is to be ethical.
- 6 a Explain the difference between the dependent and the independent variable in an experiment.
b Explain the difference between controlled and uncontrolled variables.
- 7 What is the difference between the validity and the reliability of an investigation?
- 8 Explain the difference between qualitative and quantitative data.
- 9 Describe how you would calculate the mean of a set of measurements.
- 10 What are outliers? Should outliers be excluded when drawing conclusions from a set of data?
- 11 What is a peer review? Why are peer reviews used?
- 12 Describe some of the points that should be included in the discussion section of a scientific report.
- 13 What is an 'error' when discussing a scientific investigation?

Apply your knowledge

- 1 Re-read the account of Florey's experiment in which he injected mice with penicillin (pages 3–4). What variables did Florey control in his experiment?
- 2 What did Albert Einstein mean when he said: 'No amount of experimentation can ever prove me right; a single experiment can prove me wrong'?
- 3 What type of investigation would be best for finding a solution to the following problems? Explain the reasons for your choice in each case.
 - a Can people taste the difference between two different brands of milk chocolate?
 - b What proportion of students in your school are left-handed?
 - c What is the ratio of males to females in your Human Biology class?
 - d How has a particular person's growth rate changed from birth to age 15?
- 4 In addition to physical activity that is part of their job or daily routine, many people deliberately exercise by going to a gym or by walking or jogging. Describe how you would conduct a survey to find out the average amount of time the teachers at your school spend on deliberate exercise.
- 5 The table below shows the systolic blood pressure of students in a Year 12 Human Biology class.

Table 1.2 Systolic blood pressures of Year 12 students

Systolic blood pressure (mmHg)					
109	123	141	115	131	126
144	138	106	115	49	109
125	132	128	114	116	120
195	143	132	116	113	